

ADAPTIVE DIGITAL TWIN-DRIVEN PLANNING FOR MULTI-ROBOT NAVIGATION IN GPS-DENIED ENVIRONMENTS



Mentored By:
Dr Om Prakash Pahari



Badrikanath Praharaj
22BAC10032



MV Sreeram
22BAC10004

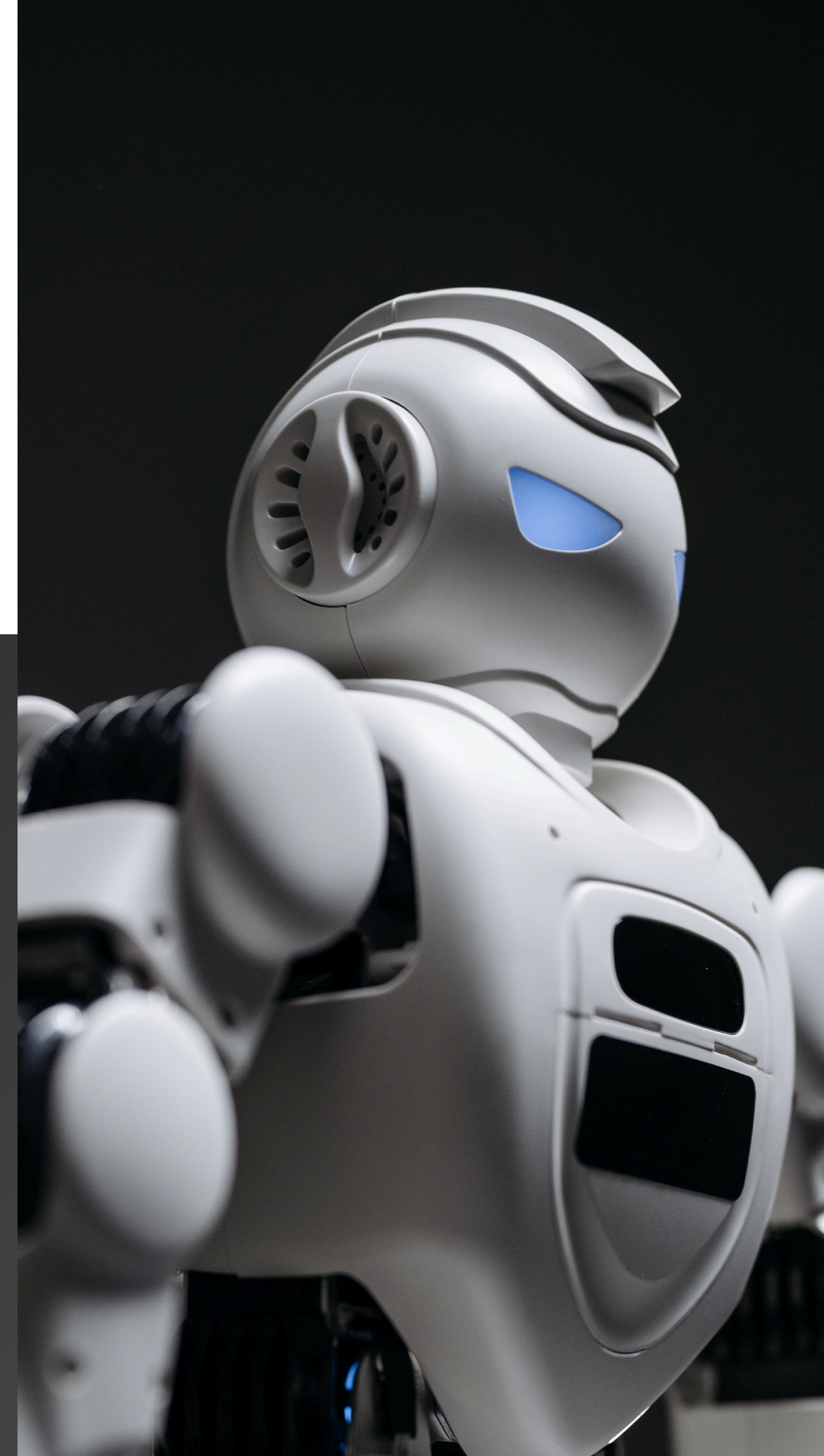
INTRODUCTION

Digital Twin

- A Digital Twin is a real-time virtual model of a robot that mirrors its state, movements, and sensor readings. It allows continuous tracking, prediction, and adaptation of the robot's behavior in complex environments.

GPS-Denied Navigation

- In tunnels, underground mines, or disaster-hit zones where GPS signals fail, Digital Twins provide reliable situational awareness. This ensures robots can still navigate safely and complete their missions.



PROPOSED APPROACH

Adaptive Digital Twin

The twin is adaptive, meaning it updates itself using real-time sensor inputs. This keeps the virtual model synchronized with the physical robot, even in changing and unpredictable environments.

Event-Triggered Planning

Instead of constant updates, the system only triggers planning when needed (like detecting an obstacle). This reduces communication overhead, saves energy, and ensures smooth multi-robot navigation.





SYSTEM ARCHITECTURE

ROS 2 Framework

ROS 2 serves as the backbone, enabling communication between sensors, control, and planning modules. It ensures multiple robots and their Digital Twins can share data and coordinate effectively.

Gazebo & RViz

Gazebo simulates the 3D world where robots operate, while RViz visualizes sensor data and robot movement. Together, they allow testing, teleoperation, and validation before deploying robots in the real world.

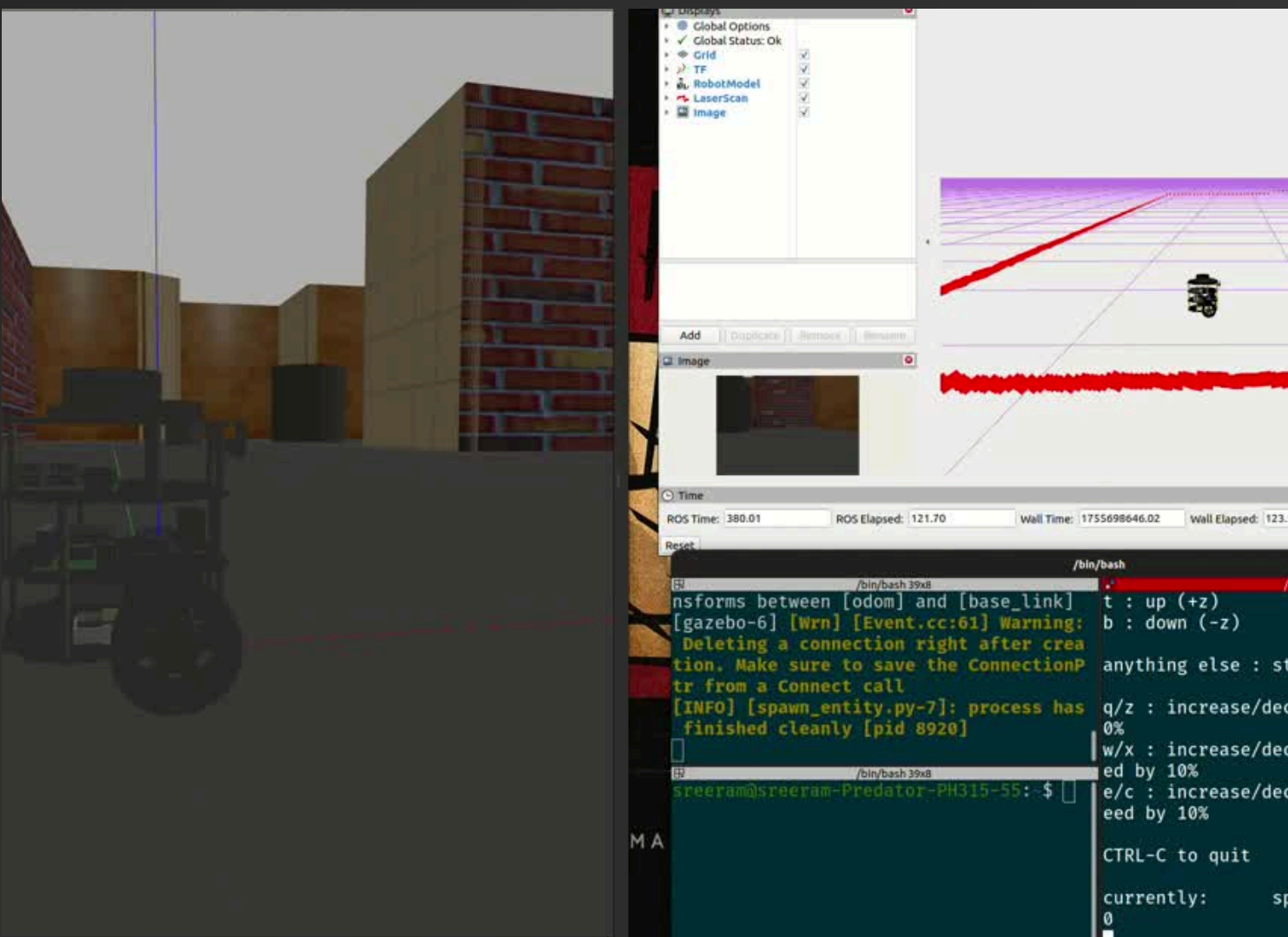
Video 1: Sourcing ROS 2 Workspace & Launching Bring-Up Files

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greeram@greeram-Predator-PH315-55:~$ source /opt/ros/foxy/setup.bash
greeram@greeram-Predator-PH315-55:~$ ros2 launch my_robot bringup.launch
```

This video shows how the ROS 2 workspace is sourced to set up the environment for robot control. The bring-up files are launched to start essential nodes and load configurations. The URDF/XML file defines the robot's physical structure and sensors, making it ready for simulation and navigation.

Video 2: Teleop Control & Visualization in RViz and Gazebo



This video demonstrates teleoperation commands to manually control the robot. The robot's motion is visualized in RViz, while Gazebo simulates its behavior in a realistic 3D world. Together, they provide both command feedback and environment interaction, ensuring accurate testing of robot navigation.

REAL-WORLD APPLICATIONS



Disaster & Defense

Robots guided by Digital Twins can explore collapsed sites or GPS-jammed zones in defense operations. They reduce risks to human lives and ensure safe exploration in hazardous environments.

Industry & Logistics

In factories and warehouses, fleets of robots can use Digital Twins to coordinate tasks. This improves productivity, reduces collisions, and enables efficient goods transport without GPS reliance.

VIT BHOPAL

THANKYOU